

location

April 3, 2025

1 Example Testing

1.1 Package

Declare the package path contained:

```
[77]: import sys
      sys.path.append('ipynb/location/packages')
```

1.1.1 Graph Visualization

Source referred: [NetworkX](#)

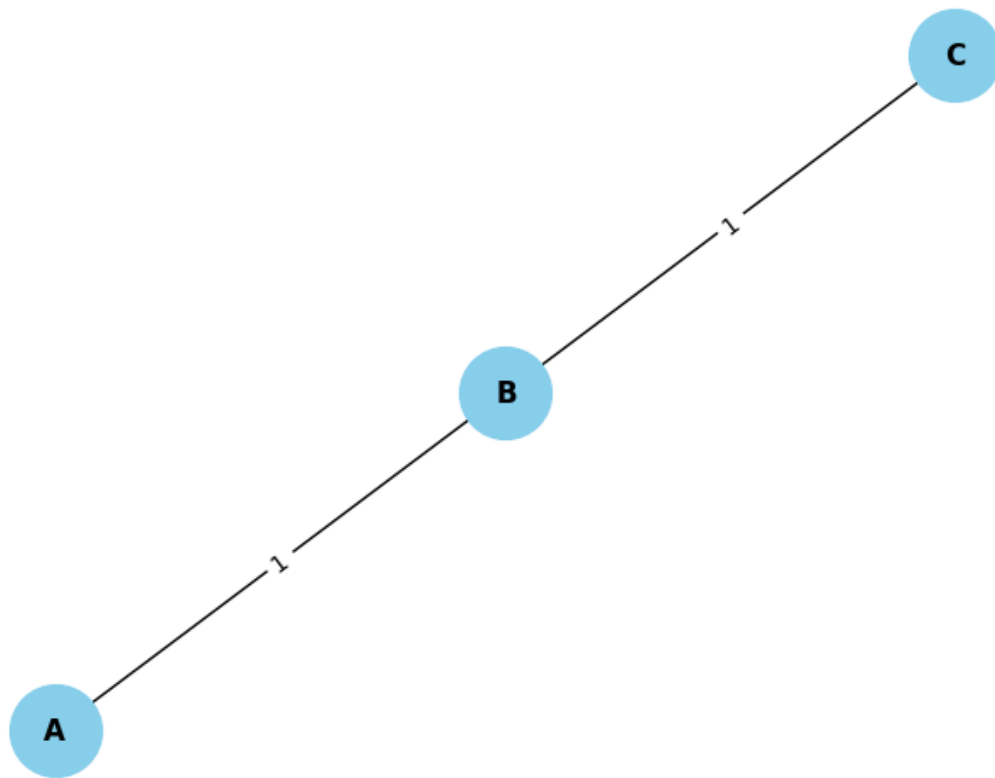
Declaring

```
[78]: from packages.GraphView import Graph
```

Example

```
[79]: matrix = [
      [0, 1, 0],
      [1, 0, 1],
      [0, 1, 0]
      ]

      G = Graph(matrix)
      G.show()
```



1.1.2 Shortest-path

Source referred: [GeeksforGeeks](#)

Declaring

```
[80]: from packages import ShortestPath
```

Example

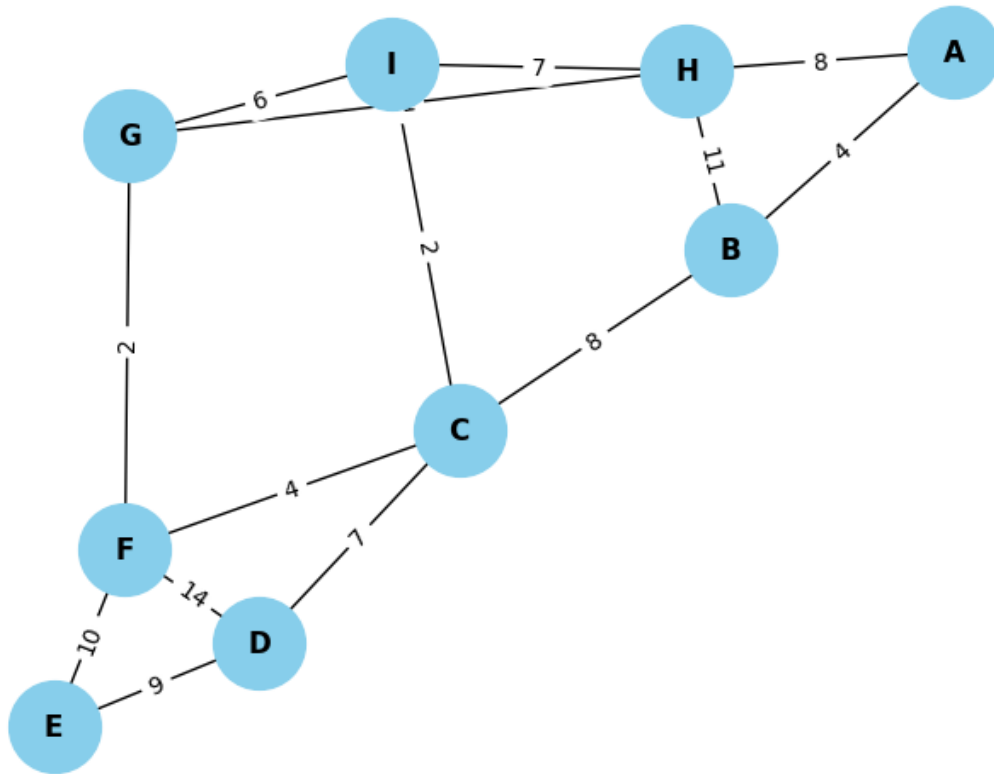
Graph parameter

```
[81]: g = ShortestPath.Graph(9)
      g.graph = [[0, 4, 0, 0, 0, 0, 0, 8, 0],
                  [4, 0, 8, 0, 0, 0, 0, 11, 0],
                  [0, 8, 0, 7, 0, 4, 0, 0, 2],
                  [0, 0, 7, 0, 9, 14, 0, 0, 0],
                  [0, 0, 0, 9, 0, 10, 0, 0, 0],
                  [0, 0, 4, 14, 10, 0, 2, 0, 0],
                  [0, 0, 0, 0, 0, 2, 0, 1, 6],
                  [8, 11, 0, 0, 0, 0, 1, 0, 7],
                  [0, 0, 2, 0, 0, 0, 6, 7, 0]]
```

```
]
```

Visualization

```
[82]: Visualization = Graph(g.graph)
Visualization.show()
```



Input 2 point requested

```
[83]: start = 'D'
      end = 'A'
```

Execute

```
[84]: startInput = ord(start) - 65
      endInput = ord(end) - 65
      path, distance = g.getShortestPath(startInput, endInput)
      result = [chr(65 + i) for i in path]
      print(result)
      print("Total distance:", distance)
```

```
['D', 'C', 'B', 'A']
```

```
Total distance: 19
```

1.2 Bottleneck point

1.2.1 Description

For $v \in V$, $x = (e, \alpha) \in G$, the bottleneck point v on $e = (v_i, v_j)$ is

$$\alpha l_e + d(v_i, v) = (1 - \alpha) l_e + d(v_j, v)$$

- BP_e : Set of all bottleneck points on $e \in E$.
- BP : Set of all bottleneck points on G .

1.2.2 Running

Declaring

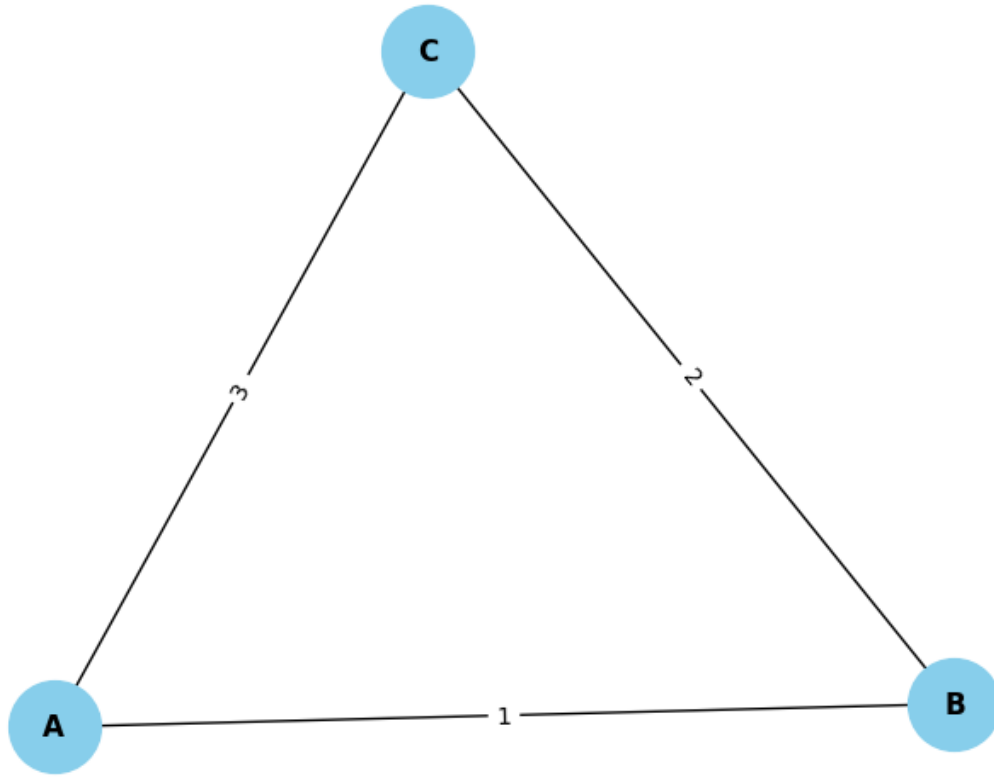
```
[85]: from packages.Bottleneck import BP
```

Graph

```
[86]: graph = [  
    [0, 1, 3],  
    [1, 0, 2],  
    [3, 2, 0]  
]
```

Show the graph

```
[87]: ShowGraph = Graph(graph)  
ShowGraph.show()
```



Executing

```
[88]: a = BP(graph, 3, 'A', 'C', 'B')
      print(a.alpha())
```

[2/3]

1.3 Equilibrium point

1.3.1 Description

For $v_k, v_l \in V$, $v_k \neq v_l$, $x = (e, \alpha) \in G$, $e = (v_u, v_v)$, the equilibrium point is

$$\omega_k(\alpha l_e + d(v_k, v_u)) = \omega_l((1 - \alpha)l_e + d(v_l, v_v))$$

If $x \in G$

$$\omega_k d(v_k, x) = \omega_l d(v_l, x)$$

- EQ_e : Set of all equilibrium points on $e \in E$.
- EQ : Set of all equilibrium points on G .

1.4 Local center

1.4.1 Description

For $e \in E$, $e = (v_i, v_j)$, the local center on e is

$$\arg \min_{x \in e} f(x) = \arg \min \{ \min_{x \in EQ_e} f(x), f(v_i), f(v_j) \}$$

1.5 Global center